



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

- Chapter 3 – Manufacturing Engineering

Manufacturing Engineering

- Designing the production process for a product
 - Everything related to the *process* of making a product, such as:
 - Evaluating the manufacturability of the product
 - Process selection and parameters
 - Cutting tool material, size, and shape
 - Cutting speed
 - Depth of cut
 - Designing work-holding devices (jigs and fixtures)
 - To control and secure the position of the work piece or to guide the machine on the part
 - Cost estimation
 - Quality of the part produced

Product - Production Design Interaction

- Product design: develop and evaluate the ability of the part to perform its function
 - Size, shape, strength, etc. using knowledge of physics, strength of material, etc.
- Manufacturing engineering: develop and evaluate the cost of producing the parts
 - Limitation and cost of various alternatives
- Interaction between product designers and manufacturing engineer from the very beginning
 - Designer opts for tight tolerances → high processing cost
 - The manufacturing engineer opts for the largest possible tolerance

Product - Production Design Interaction

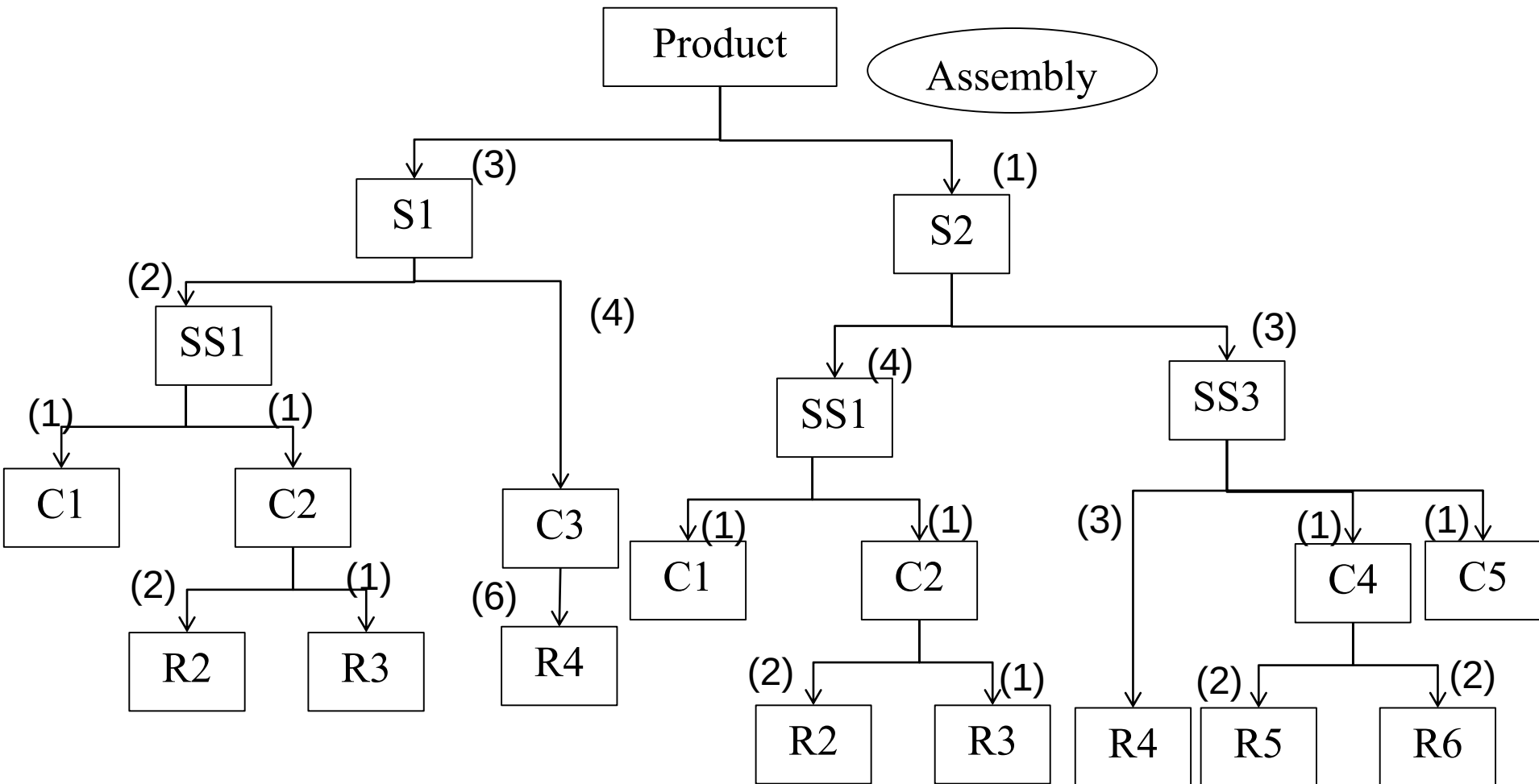
- Concurrent (simultaneous) Engineering:
 - All elements of a product's life-cycle, from functionality, producibility, assembly, testability, maintenance issues, environmental impact and finally disposal and recycling, should be taken into careful consideration in the early design phases
 - Functions of design engineering and manufacturing engineering (and other functions) are integrated to reduce the elapsed time required to bring a new product to the market
 - The integration of key functions at the outset of a process in order to emphasize the simultaneous development of a product and its associated processes
 - Has been applied most notably in the aerospace industry

Process Engineering

- Design of the actual process to be used in the manufacture of the product
- Defining product structure and specifications
- Component manufacturability and available processes
- Evaluating the cost of each alternative process
- Operations sequence
- Documentation

Product structure and specifications

- Bill of material



BOM

BILL OF MATERIAL			
Product Description: Bookcase, metal, 3 shelves. Stock Number: 1			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
3	Shelf	3	Manufacturing
4	Leg	4	Manufacturing
5	Inserts	8	Purchasing
6	Screws	12	Purchasing

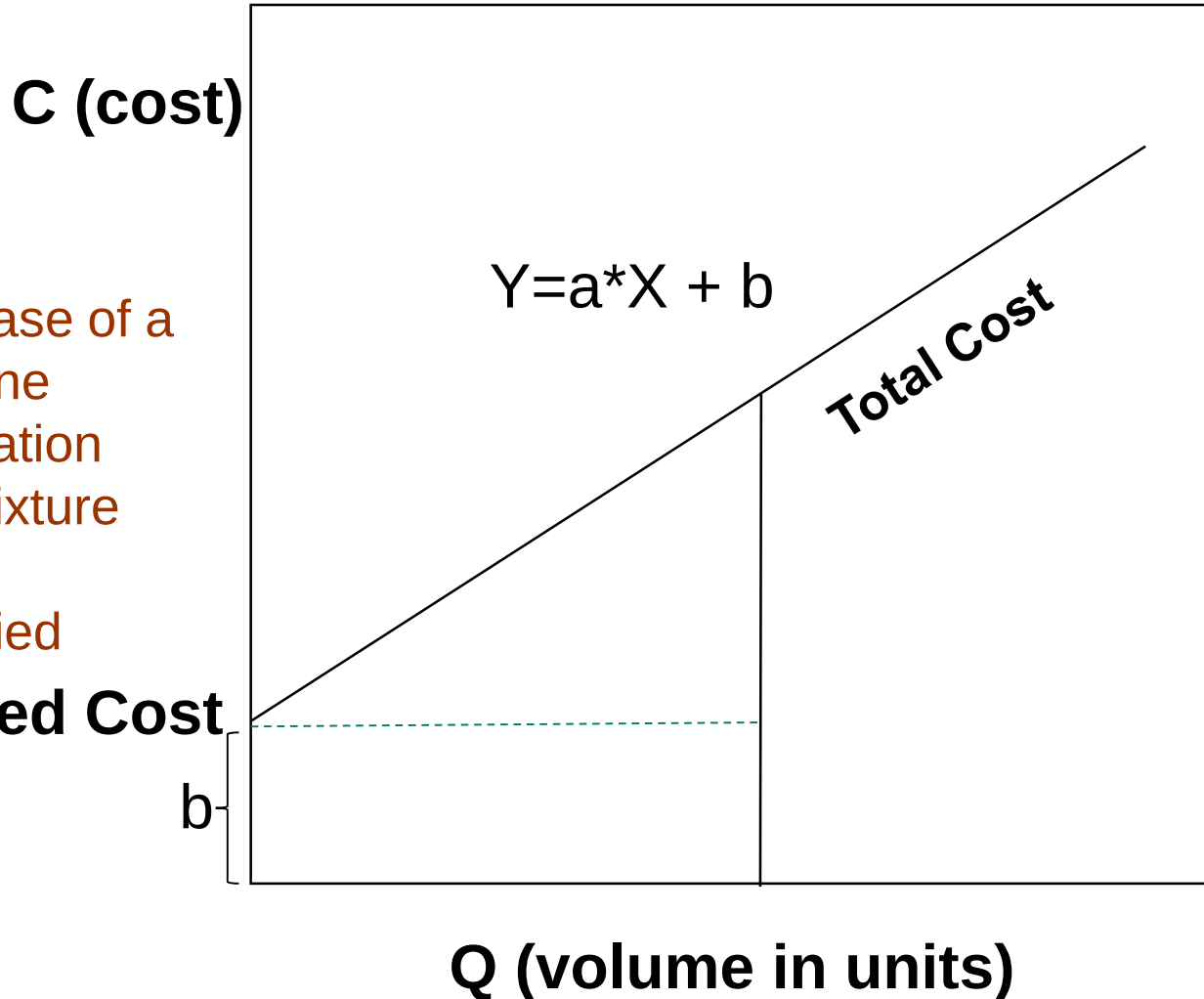
BILL OF MATERIAL			
Product Description: Shelf Stock Number: 3			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
8	Sheet metal	3 square feet	Purchasing

BILL OF MATERIAL			
Product Description: Leg Stock Number: 4			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
8	Sheet metal	2 square feet	Purchasing

Process Engineering

- Component manufacturability
 - Relative ease or difficulty of producing the component within tolerance
 - Tighter tolerances more expensive
 - Standardized features (e.g., size of a hole vs. commonly available drill size)
 - Commonly available material vs particular characteristics
- Available processes
 - List of candidate processes for each item
 - Compatibility of material
 - The ability to produce the desired tolerance
 - Part (re)design required to make it easy to produce

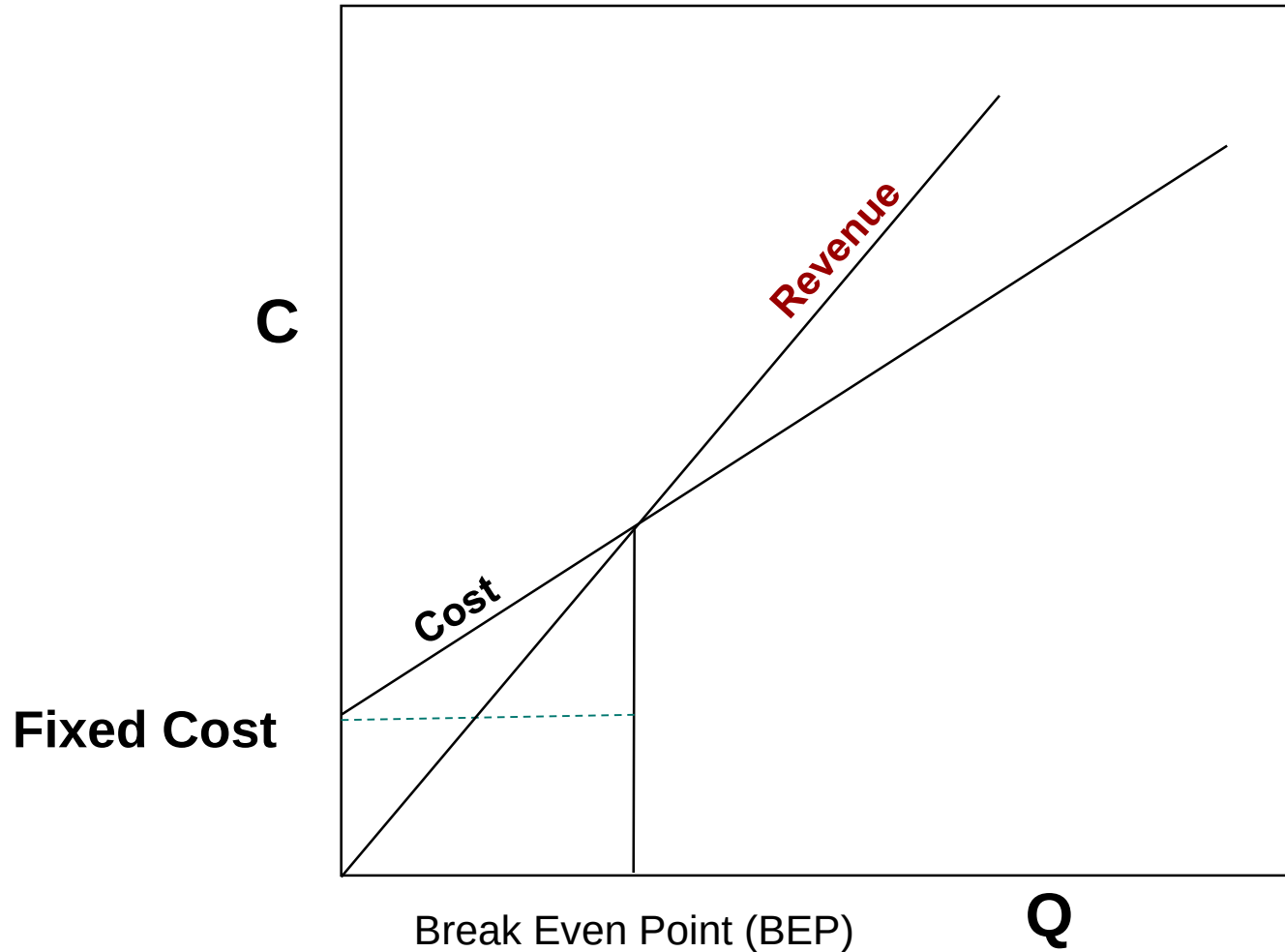
Cost-Volume Relationships



- Purchase of a machine
- Installation
- Jigs/Fixture
- Space occupied

Variable cost:
material,
labor, tools

Revenue-based Break Even Point



BEP: Level of sales at which total revenue equals total cost.

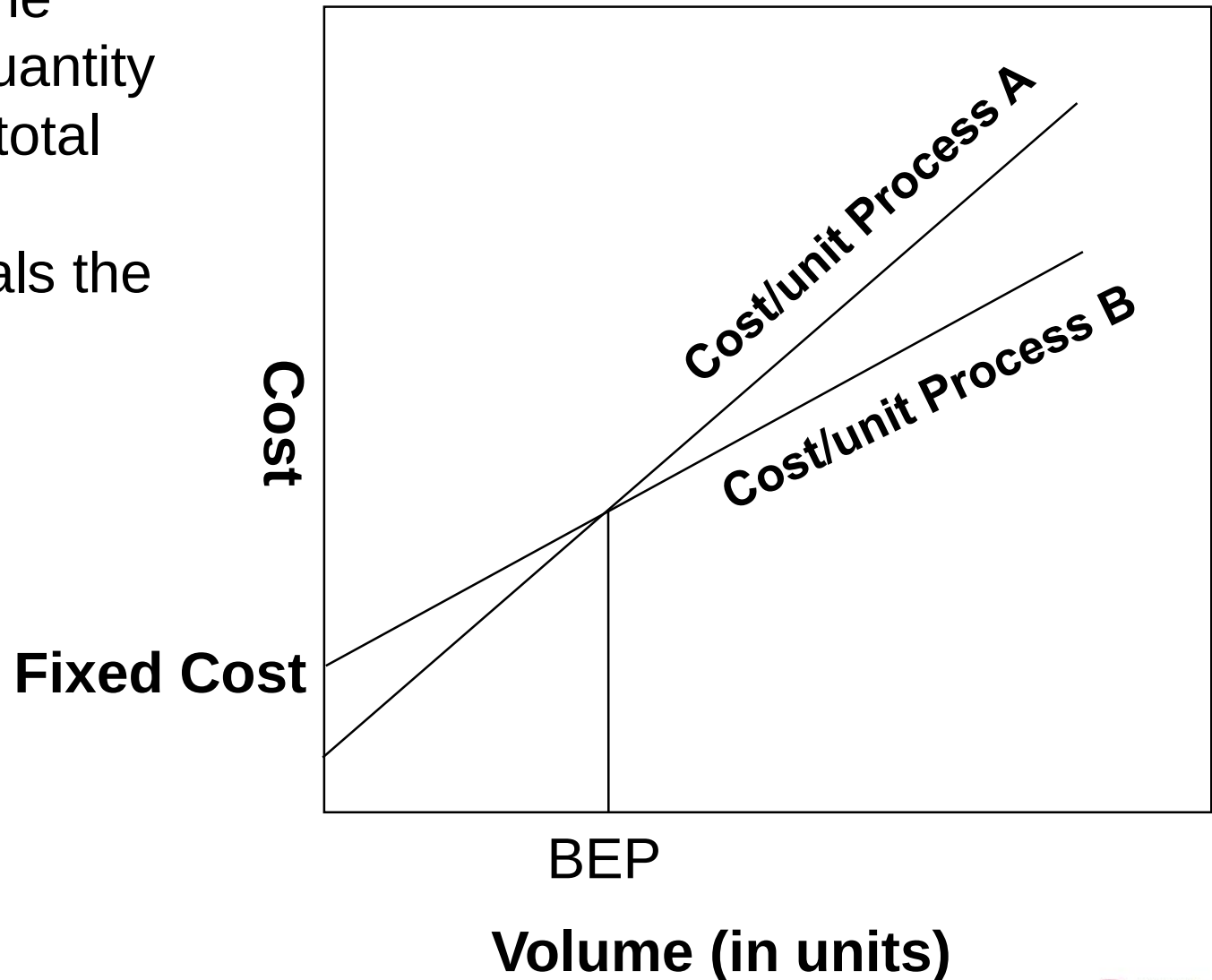
Example

- Fixed cost = \$28,000,
- Variable cost = \$100/unit
- Price per product = \$200
- Linear cost function: $Y = a \cdot X + b$
- Break-even point is X , then
$$200 \cdot X > 100 \cdot X + 28,000 \quad \text{or} \quad X > 28,000 / 100 = 280$$
- If estimated total demand is less than 280, the company should not set up the system to produce the products as fixed cost cannot be recovered.

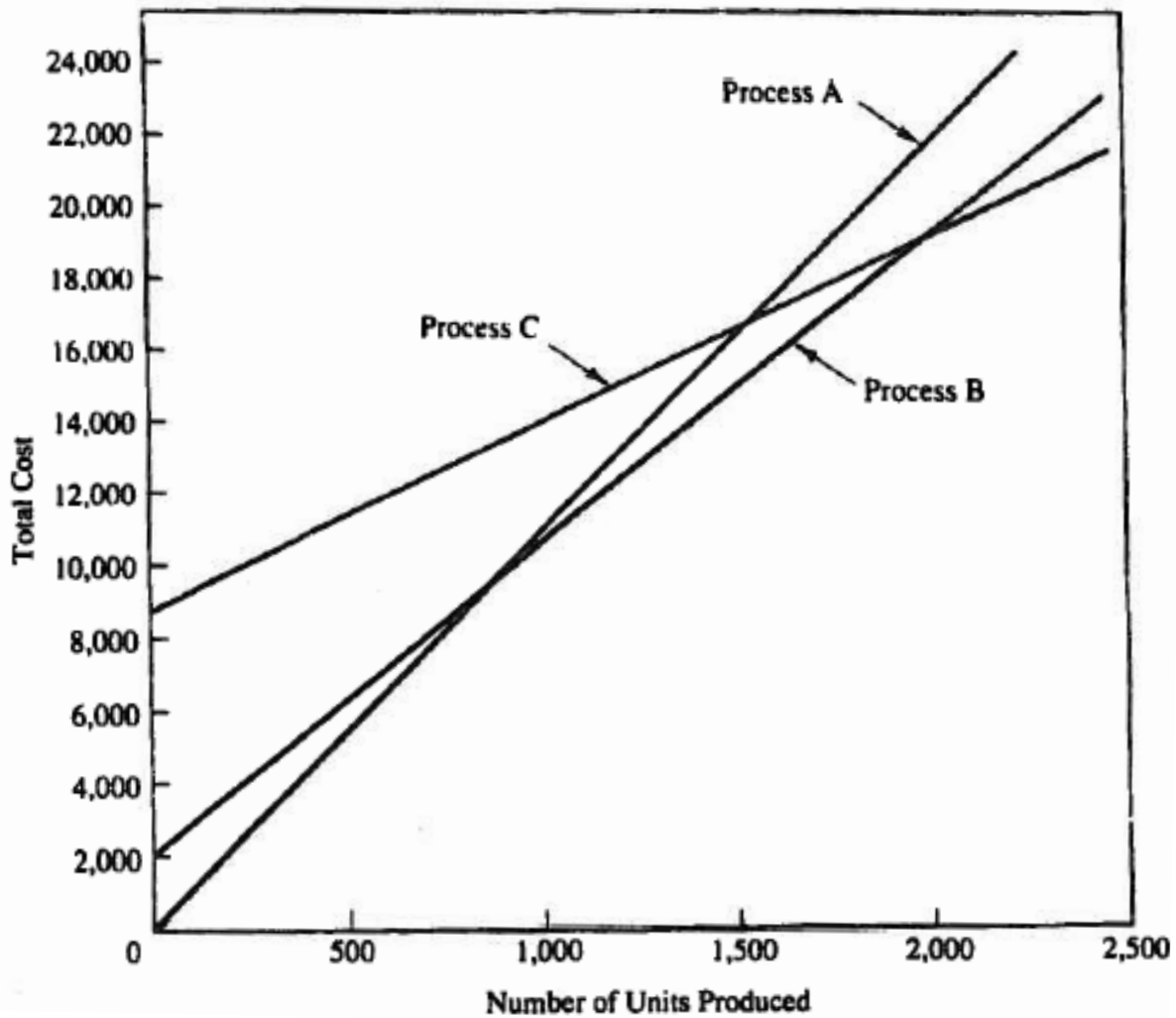


Evaluating the cost of each process

- The BEP is the production quantity at which the total cost of one process equals the total cost of another.



BEP



Example

Process	Total Annual Fixed Costs	Variable cost per unit
A	\$110,000	\$2
B	\$80,000	\$4
C	\$75,000	\$5

- a) If projected annual demand is 10,000 units per year, which process should be selected?
- b) At what volume would each of these processes be preferred?

Illustrative example



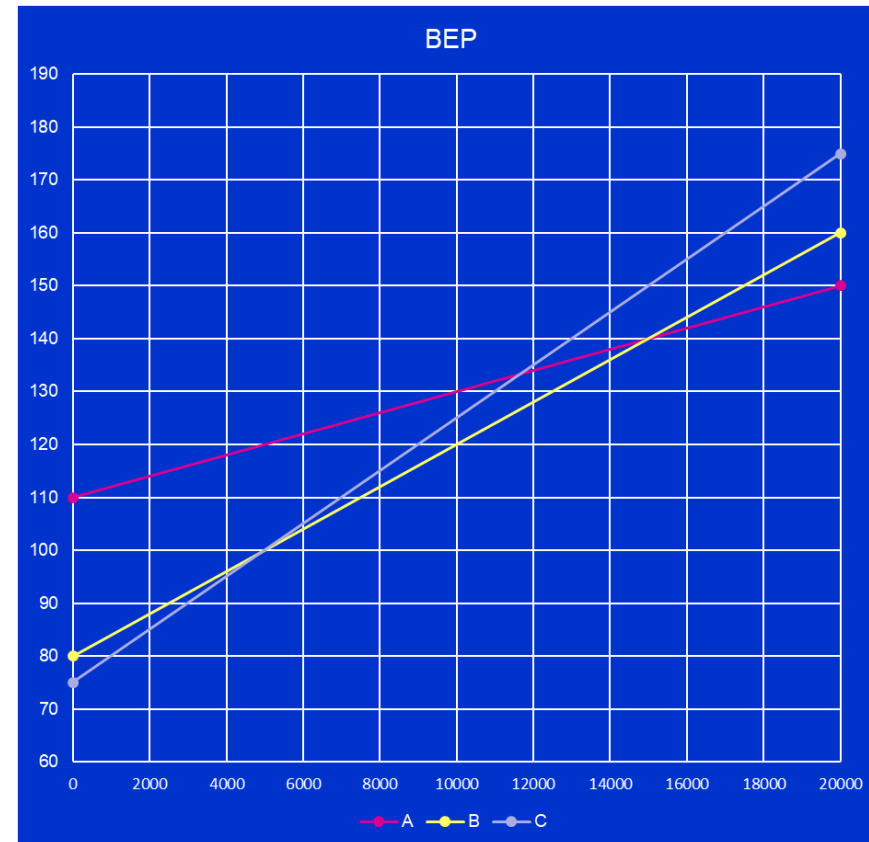
- Metal bracket
- Alternative processes
 - CNC machining
 - High purchasing cost
 - High production cost (slow cycle time-high precision)
 - Metal stamping and bending
 - High set-up cost (dies, tooling)
 - Low production cost
 - 3D printing
 - High purchasing cost
 - High production cost (slow cycle time)

Solution

Linear cost function

$$Y = a * X + b$$

- a)
 - $TC(A) = \$130,000$
 - $TC(B) = \$120,000$
 - $TC(C) = \$125,000$
- b)
 - $BEP(BC) = 5,000$ units
 - $BEP(AB) = 15,000$ units
 - Between 0-5,000, choose C
 - Between 5,000-15,000, choose B
 - Above 15,000, choose A
 - At 15,000, either A or B
- Selling price must be \$12/unit and higher



Determining the sequence of operations

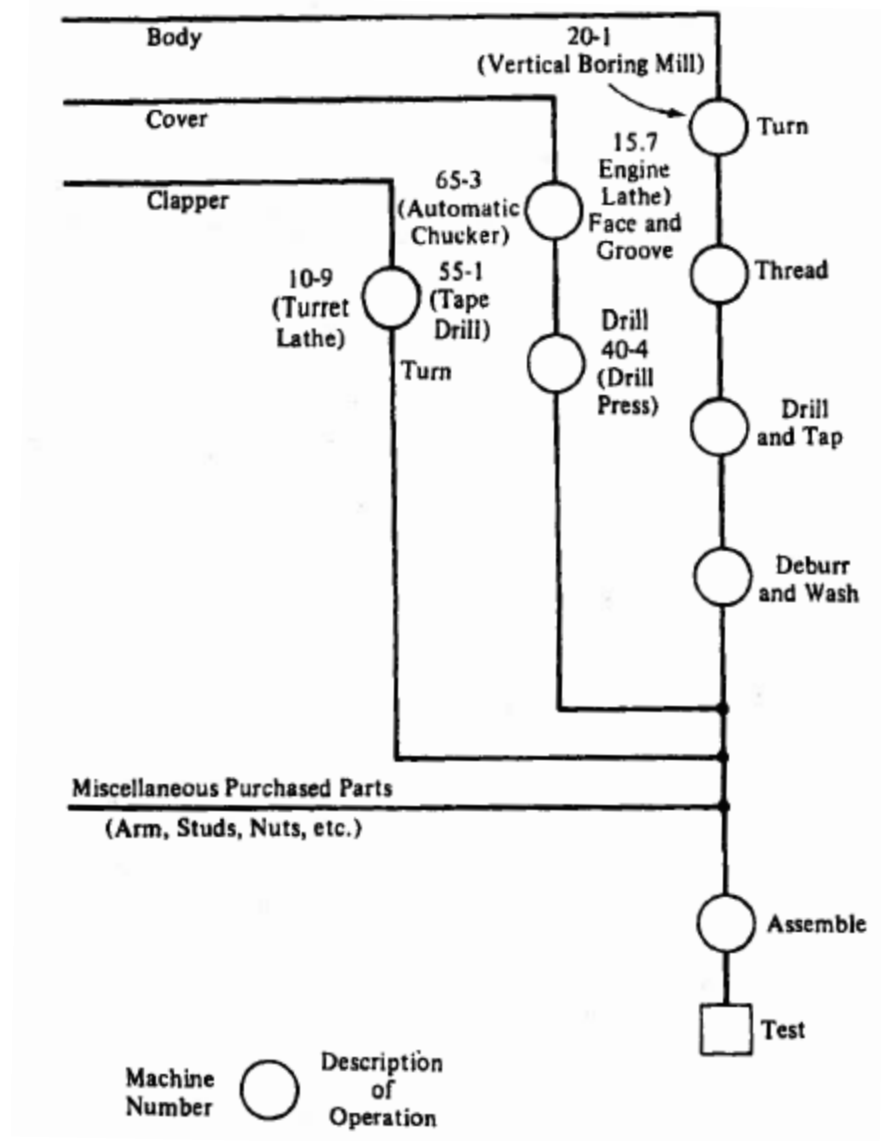
- Minimization of material handling
 - Part is routed along the shortest path without backtracking
- No succeeding operation adversely affects previous operations
 - Leaving burrs, debris, scratches, etc.
- Performing as many operations on each machine as possible
 - Close tolerance is achieved and quality assured

Determining the sequence of operations

- Sequence of Operations for a Steel Shaft
 - Cutting Stock (bandsaw)
 - Facing (Lathe)
 - Turning (Lathe)
 - Drilling (Lathe or Drill Press)
 - Grooving (Lathe)
 - Heat Treatment
 - Grinding (Cylindrical Grinder)
 - Surface Finishing
 - Apply protective coating to prevent corrosion

Determining the sequence of operations

- Operations process chart for check valve



Industrial Processes

- Refining and Alloying
- Casting
- Metal Forming
- Metal Cutting
- Welding
- Assembly
- Finishing

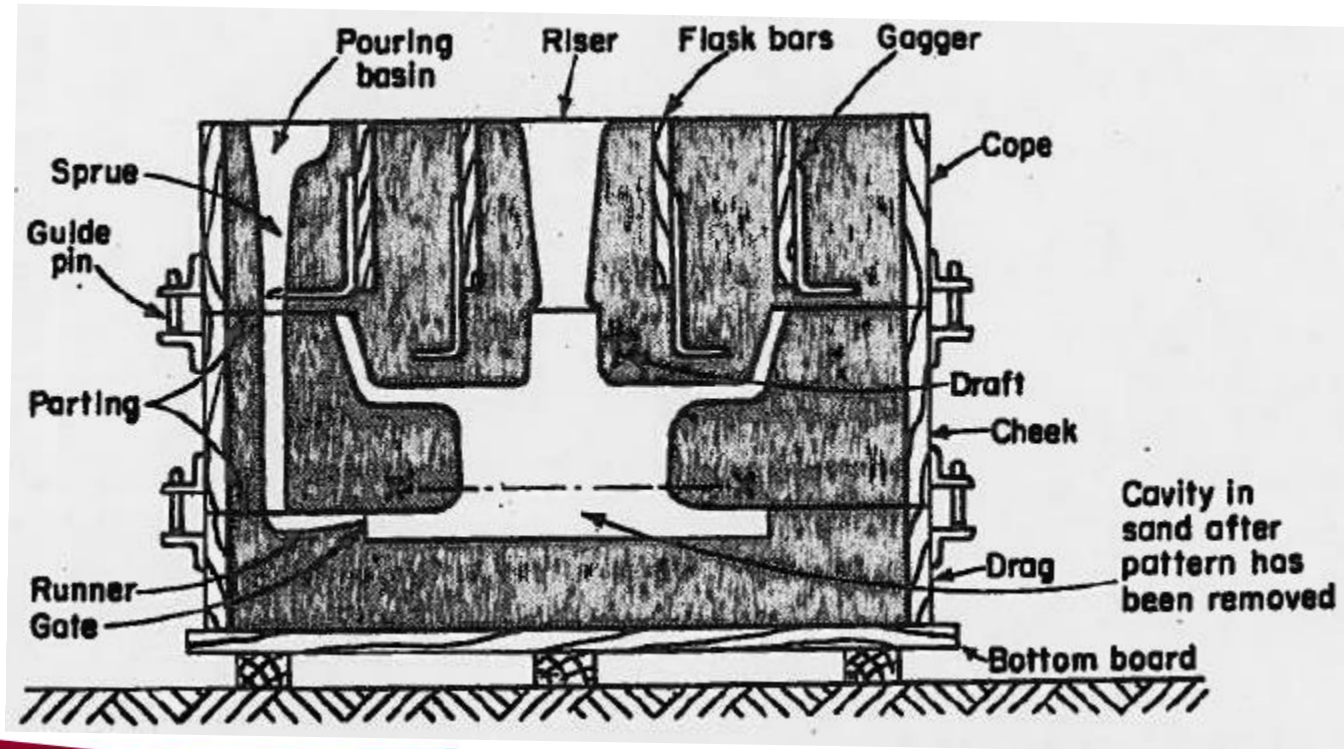
3D printing (Additive manufacturing)

Refining and Alloying

- Refining
 - Metal ore → improve its usefulness
 - Iron ore → steel
 - Blast furnace → Steel mill (open hearth, basic oxygen, electric furnace process)
 - Different types of steel depending on the temperature and chemical composition of the furnace
- Alloying
 - Metals from primary operations do not have the properties necessary for the specific design
 - Metallurgical transformation to improve the hardness, strength, workability, etc.
 - Heat treating
 - Combining metals (alloys)

Casting

- To obtain an approximate shape from a metal
- Forming objects by pouring liquid metal into a mold and allowing the metal to solidify
- Mold design is critical
- Sand casting → expensive for high production volume
- Permanente molds

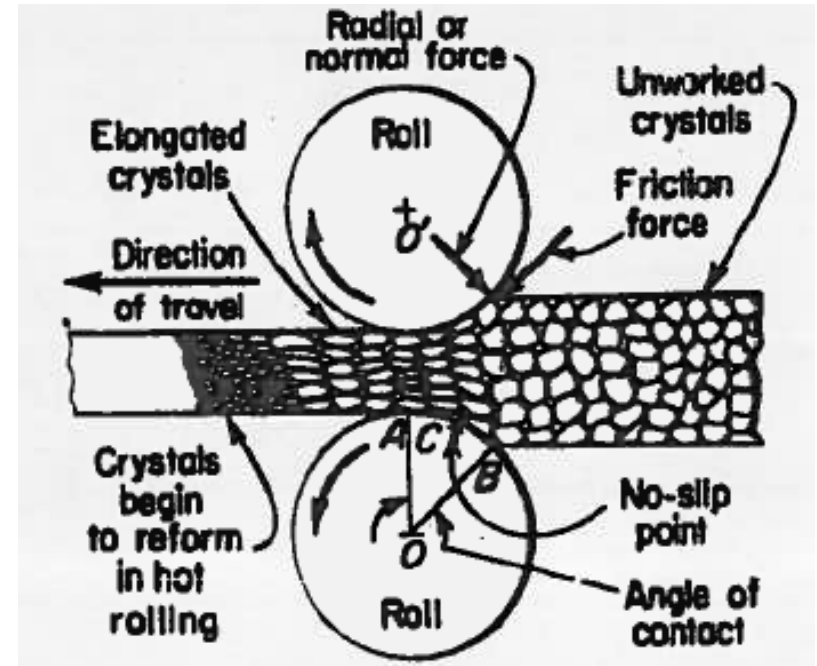


Metal forming

- The metal is worked under pressure to form a desired shape and/or to improve its physical properties (hot or cold)
- Hot working: temperature above crystallization → unusual shapes
- Cold working: hold close tolerance, good surface finish
- Rolling
- Wire drawing
- Forging
- Extrusion
- Bending
- Drawing and stretching

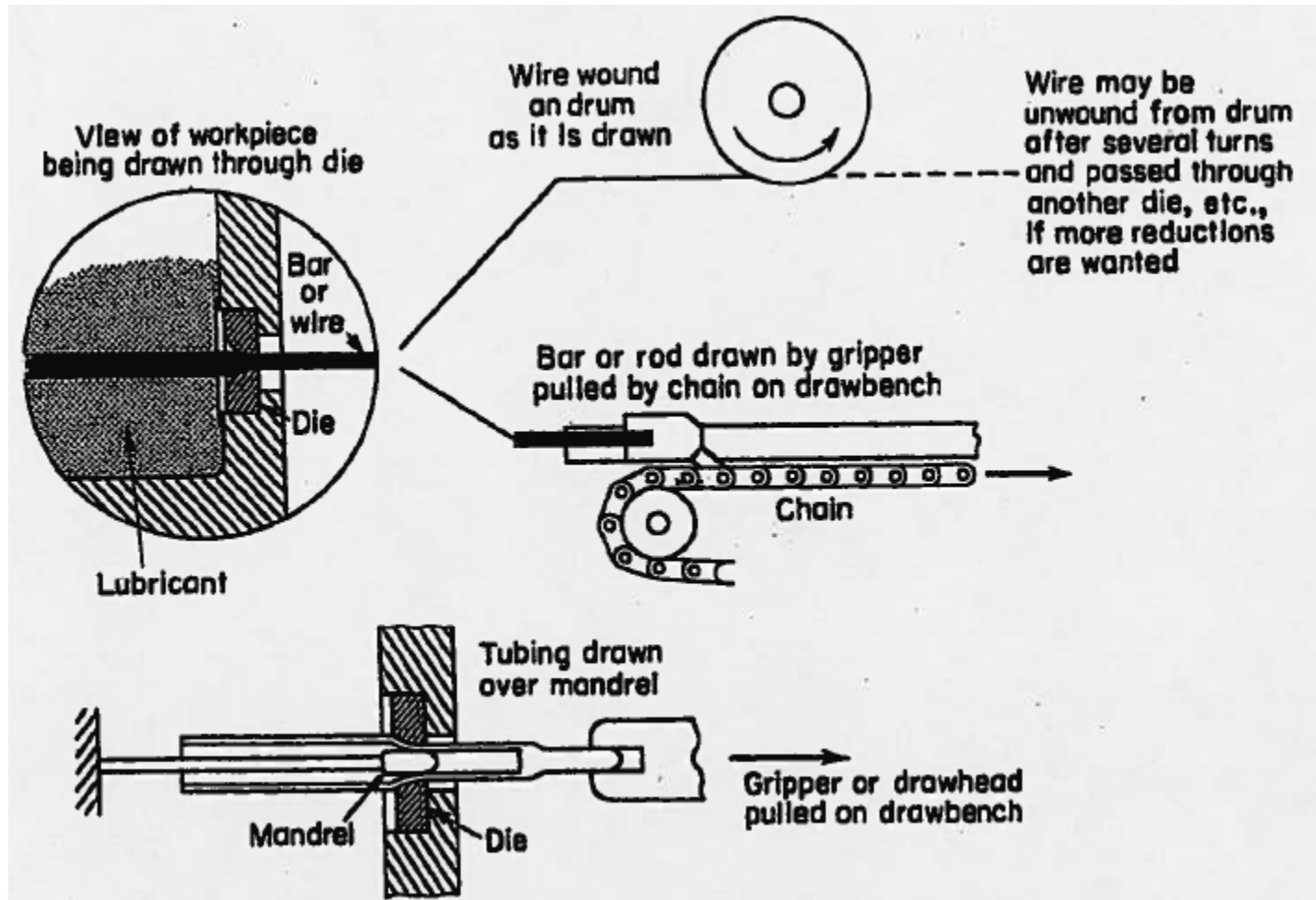
Rolling operation

- Metal is elongated or widened
 - Hot rolling (heated above its recrystallization temperature)
 - Cold rolling



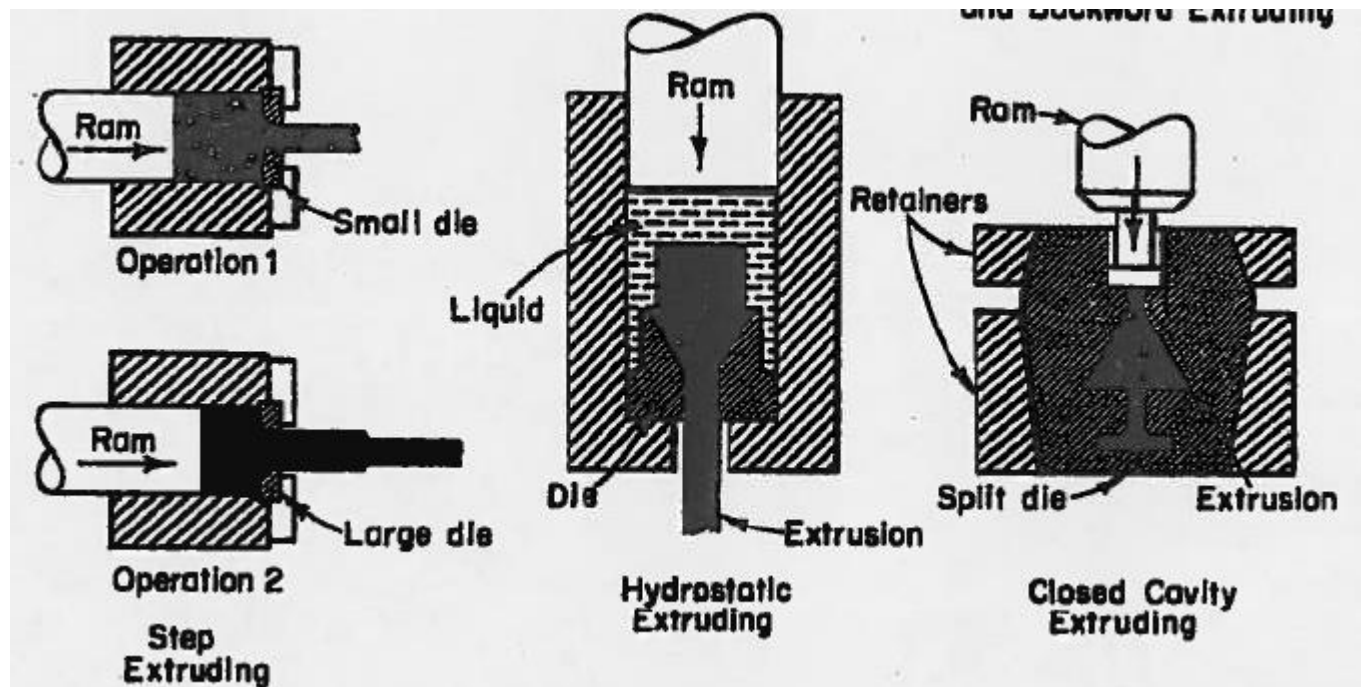
Wire drawing

Wire or rod is reduced in cross-sectional diameter by being drawn through a die



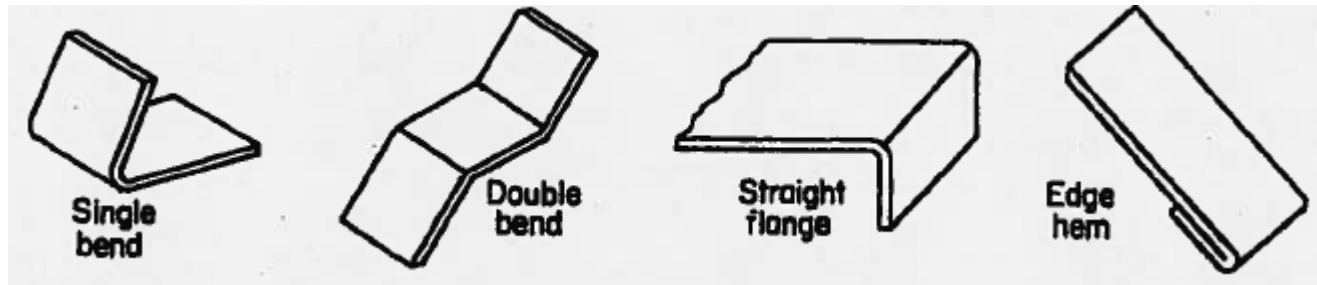
Metal forming

- Forging: a single or a series of intermittent applications of pressure (instead of continuous pressure)
 - Hammering of horseshoe
- Extrusion
 - Compressing a metal beyond its elastic limits and forcing it to flow through and to take on the shape of an opening

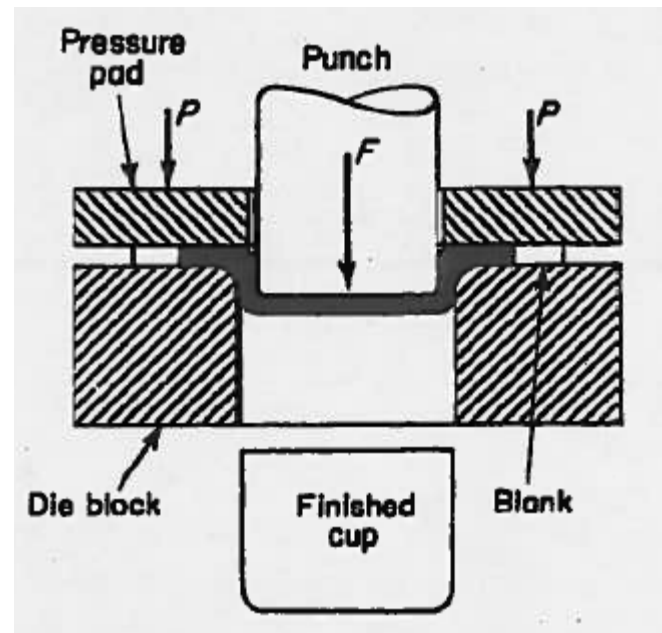


Metal forming

- Bending
 - Applying force to permanently distort the metal to the preconceived shape

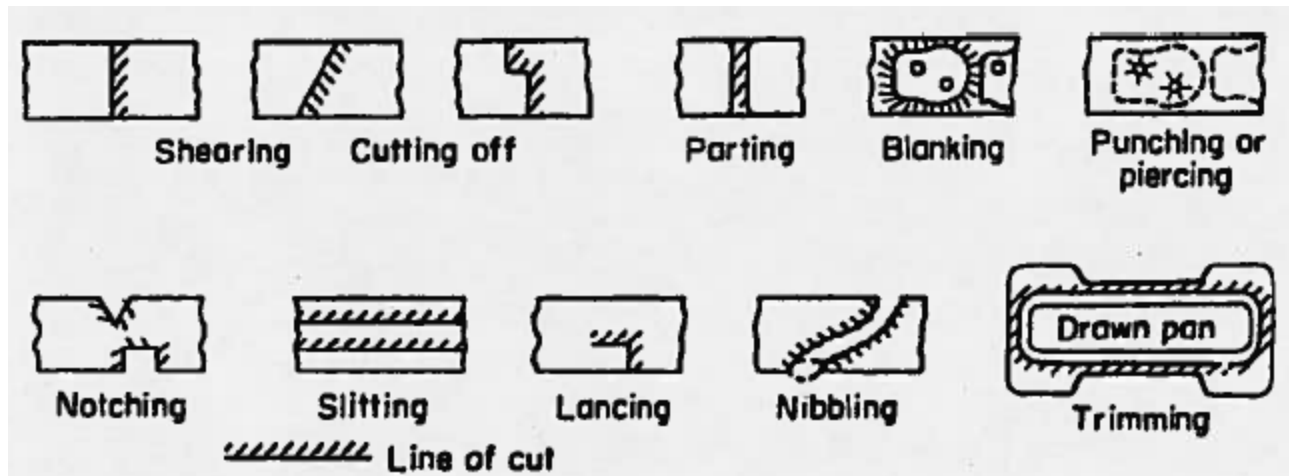


- Drawing and stretching
 - Produce seamless vessels by applying pressure and forcing the sheet metal to take on a shape



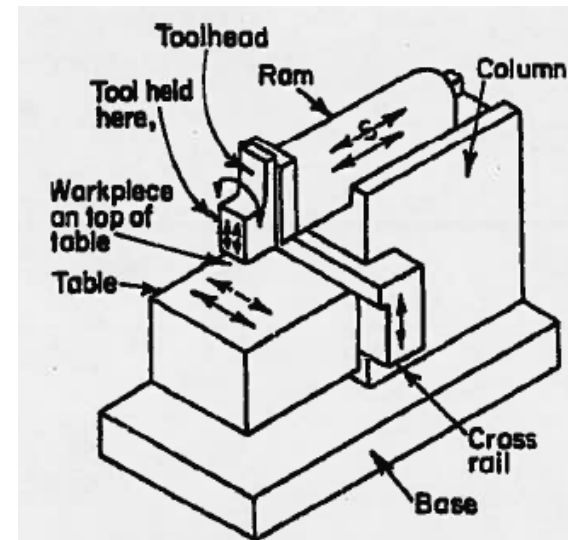
Metal cutting

- Shearing
 - Applying pressure and forcing the metal between two sharp edges
 - Blanking, parting, punching, nibbling, shearing along a straight line



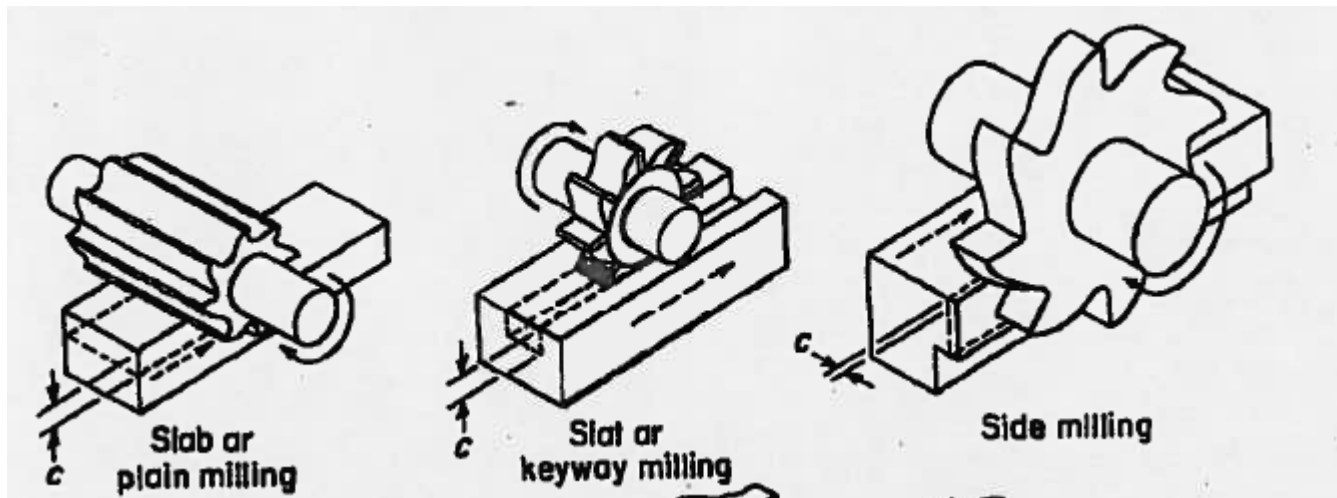
Metal cutting

- Turning
 - Rotating the workpiece and cutting metal by feeding a cutting tool
- Drilling
 - Opening, enlarging or finishing a hole
- Shaping and planing
 - Surfaces are cut by a reciprocating action
 - Shaping: the workpiece is stationary and the tool reciprocates
 - Planing: the tool is stationary



Metal cutting

- Milling
 - A revolving cutting tool with a number of teeth takes intermittent, successive cuts

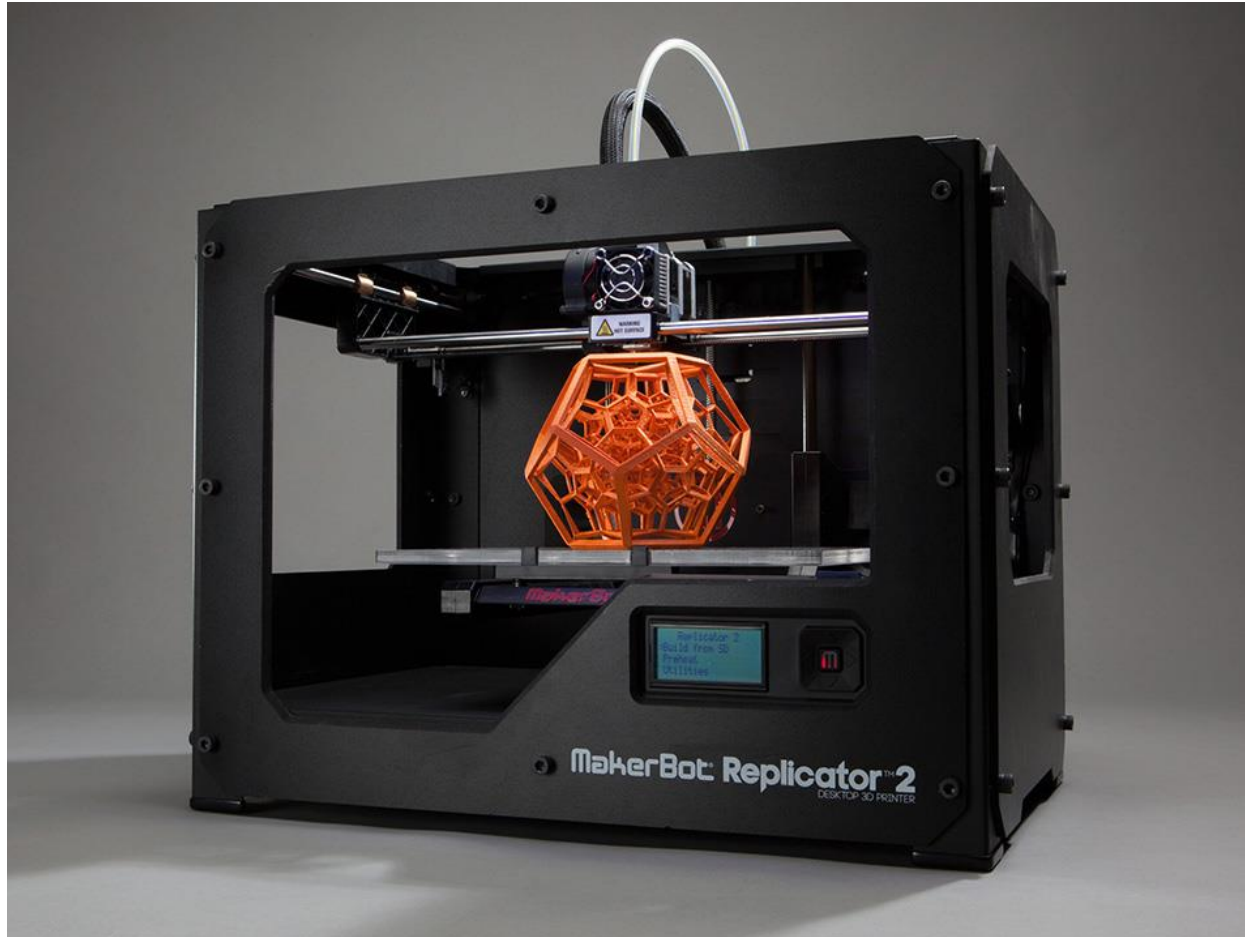


- Broaching (similar to milling- cutting tool is not revolved; it is pushed or pulled)
- Sawing-Filling
- Grinding (removal of metal in small pieces to improve surface finishes on very hard metal)

Welding

- Two pieces of the same metal are made to bond through the application of heat or pressure or both
 - Electric arc
 - Resistance welding
 - Beam welding
 - Thermit welding
 - Pressure welding
 - Gas welding
 - Brazing and soldering

3D printer

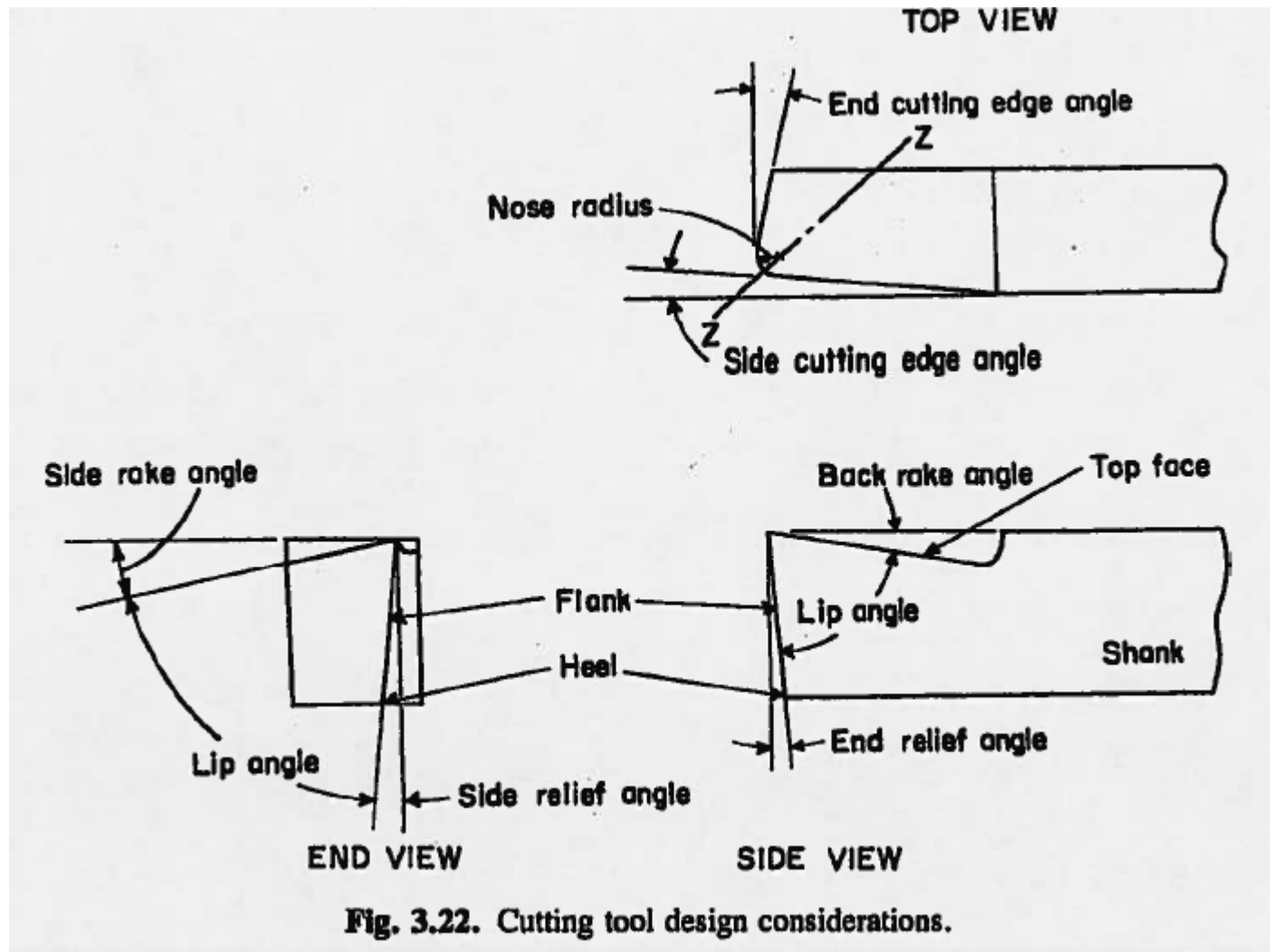


Ancillary Functions

- Tool, jig and fixture design
- Cost estimating
 - Material
 - Labor
 - Overhead (indirect costs)
- Maintenance systems design
 - Preventive maintenance for exceptionally important machines
 - Emergency maintenance (for machines that occasionally break)
- Packaging systems
 - To protect the final product during transit against damage
 - Various design options with their cost

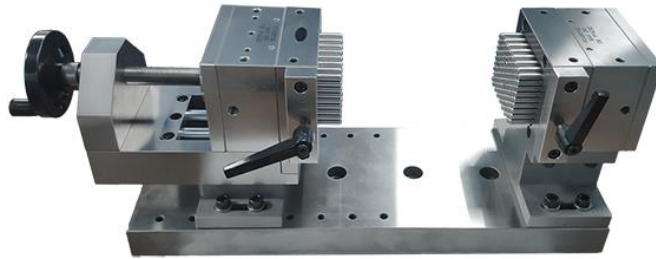
Tool design considerations

Tool design for the most effective operation



Jig and fixture design

- Fixture: a device used to hold and locate a workpiece
 - Vise



- Jig: holds the workpiece and guides the cutting tool into the workpiece
 - Drilling jig

